

Name:O.ISRAEL H.No:2303A51825 Batch:26

SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING	
Program Name: B. Tech		Assignment Type: Lab	Academic Year:2025-2026
Course Coordinator Name		Dr. Rishabh Mittal	
Instructor(s) Name		Mr. S Naresh Kumar	
		Ms. B. Swathi	
		Dr. Sasanko Shekhar Gantayat	
		Mr. Md Sallauddin	
		Dr. Mathivanan	
		Mr. Y Srikanth	
		Ms. N Shilpa	
		Dr. Rishabh Mittal (Coordinator)	
		Dr. R. Prashant Kumar	
		Mr. Ankushavali MD	
		Mr. B Viswanath	
		Ms. Sujitha Reddy	
		Ms. A. Anitha	
		Ms. M.Madhuri	
		Ms. Katherashala Swetha	
		Ms. Velpula sumalatha	
		Mr. Bingi Raju	
CourseCode	23CS002PC304	Course Title	AI Assisted Coding
Year/Sem	III/II	Regulation	R23
Date and Day of Assignment	Week3 – Wednesday	Time(s)	23CSBTB01 To 23CSBTB52
Duration	2 Hours	Applicable to Batches	All batches
Assignment Number:8.3(Present assignment number)/24(Total number of assignments)			
Q.No.	Question		Expected Time to complete
1	Lab 8: Test-Driven Development with AI – Generating and Working with Test Cases Lab Objectives • Introduce TDD using AI • Generate test cases before implementation • Emphasize testing and validation • Encourage clean, reliable code Lab Outcomes Students will be able to: • Write AI-generated test cases		Week4 - Wednesday

- Implement code using test-first approach
- Validate using unittest
- Analyze test coverage
- Compare AI vs manual tests

Task 1: Email Validation using TDD

Scenario

You are developing a user registration system that requires reliable email input validation.

Requirements

- Must contain @ and . characters
- Must not start or end with special characters
- Should not allow multiple @ symbols
- AI should generate test cases covering valid and invalid email formats
- Implement is_valid_email(email) to pass all AI-generated test cases

Expected Output

- Python function for email validation
- All AI-generated test cases pass successfully
- Invalid email formats are correctly rejected
- Valid email formats return True

```

1  def is_valid_email(email):
2      if not isinstance(email, str) or not email or email.count('@') != 1 or '.' not in email:
3          return False
4      if email[0] in '@.' or email[-1] in '@.':
5          return False
6      local, domain = email.split('@')
7      if local.startswith('.') or local.endswith('.'):
8          return False
9      if domain.startswith('.') or domain.endswith('.'):
10         return False
11     return bool(local and domain and '.' in domain)
12
13 print("user@example.com is valid:", is_valid_email("user@example.com"))
14 print("test.email@domain.co.uk is valid:", is_valid_email("test.email@domain.co.uk"))
15 print(" is invalid:", is_valid_email(""))
16 print("example.com is invalid:", is_valid_email("example.com"))
17 print("@example.com is invalid:", is_valid_email("@example.com"))
18 print("example.com@ is invalid:", is_valid_email("example.com@"))
19 print("user@@domain.com is invalid:", is_valid_email("user@@domain.com"))
20 print("user@.com is invalid:", is_valid_email("user@.com"))
21 print(".user@domain.com is invalid:", is_valid_email(".user@domain.com"))
22 print("user@domain. is invalid:", is_valid_email("user@domain."))
23 print("user@domaincom is invalid:", is_valid_email("user@domaincom"))

```

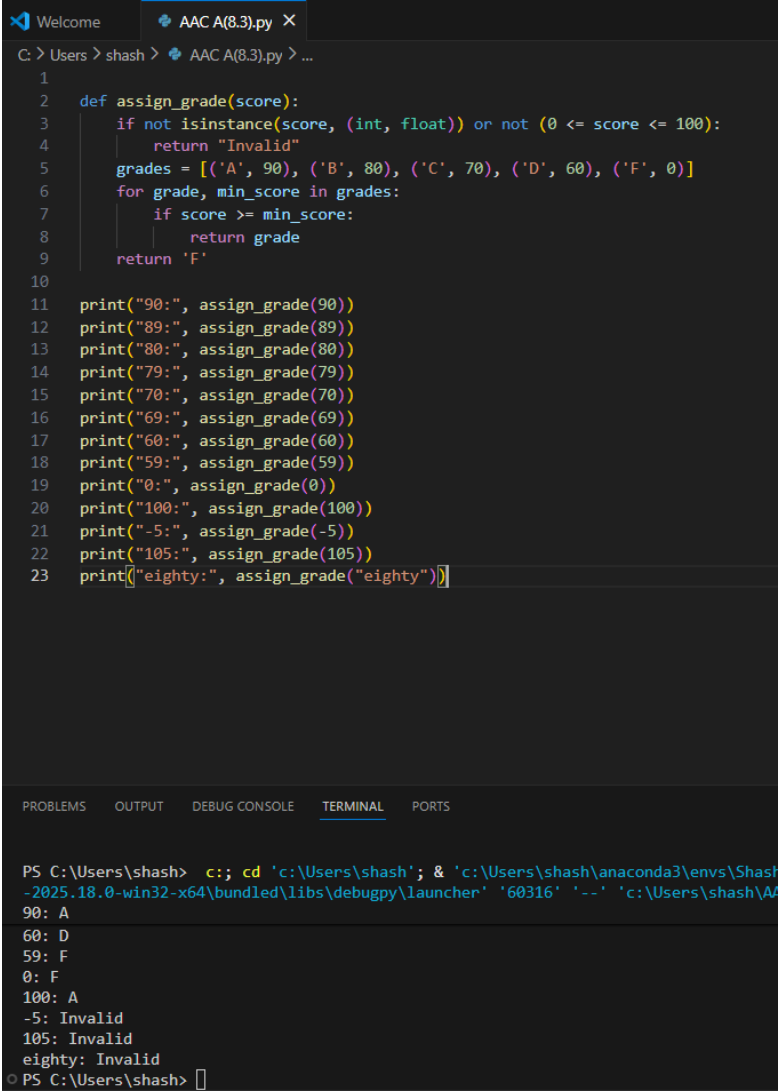
```

PS C:\Users\shash> c;; cd 'c:\Users\shash'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' '-c'
-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '58284' '--' 'c:\Users\shash\AAC A(8.3).py'
user@example.com is valid: True
@example.com is invalid: False
example.com@ is invalid: False
user@@domain.com is invalid: False
user@.com is invalid: False
.user@domain.com is invalid: False
user@domain. is invalid: False
user@domaincom is invalid: False
PS C:\Users\shash>

```

Task 2: Grade Assignment using Loops

Scenario

	You are building an automated grading system for an online examination platform. Requirements - AI should generate test cases for <code>assign_grade(score)</code> where: 90–100 → A 80–89 → B 70–79 → C 60–69 → D - Below 60 → F - Include boundary values (60, 70, 80, 90) - Include invalid inputs such as -5, 105, "eighty" - Implement the function using a test-driven approach Expected Output - Grade assignment function implemented in Python - Boundary values handled correctly - Invalid inputs handled gracefully - All AI-generated test cases pass  <pre> 1 2 def assign_grade(score): 3 if not isinstance(score, (int, float)) or not (0 <= score <= 100): 4 return "Invalid" 5 grades = [('A', 90), ('B', 80), ('C', 70), ('D', 60), ('F', 0)] 6 for grade, min_score in grades: 7 if score >= min_score: 8 return grade 9 return 'F' 10 11 print("90:", assign_grade(90)) 12 print("89:", assign_grade(89)) 13 print("80:", assign_grade(80)) 14 print("79:", assign_grade(79)) 15 print("70:", assign_grade(70)) 16 print("69:", assign_grade(69)) 17 print("60:", assign_grade(60)) 18 print("59:", assign_grade(59)) 19 print("0:", assign_grade(0)) 20 print("100:", assign_grade(100)) 21 print("-5:", assign_grade(-5)) 22 print("105:", assign_grade(105)) 23 print("eighty:", assign_grade("eighty")) </pre> <pre> PS C:\Users\shash> c:; cd 'c:\Users\shash'; & 'c:\Users\shash\anaconda3\envs\Shash-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '60316' '--' 'c:\Users\shash\A 90: A 60: D 59: F 0: F 100: A -5: Invalid 105: Invalid eighty: Invalid PS C:\Users\shash> </pre>	
	Task 3: Sentence Palindrome Checker Scenario You are developing a text-processing utility to analyze sentences. Requirements AI should generate test cases for <code>is_sentence_palindrome(sentence)</code>	

- Ignore case, spaces, and punctuation
 - Test both palindromic and non-palindromic sentences
 - Example:
– "A man a plan a canal Panama" → True
- Expected Output
- Function correctly identifies sentence palindromes
 - Case and punctuation are ignored
 - Returns True or False accurately
 - All AI-generated test cases pass

```

1  import re
2  def is_sentence_palindrome(sentence):
3      cleaned = re.sub(r'^a-zA-Z0-9', '', sentence).lower()
4      return cleaned == cleaned[::-1]
5  test_cases = [
6      ("A man a plan a canal Panama", True),
7      ("Racecar", True),
8      ("Was it a car or a cat I saw?", True),
9      ("Hello world", False),
10     ("This is not a palindrome", False),
11     (" ", True),
12     ("a", True),
13     ("A", True),
14     ("ab", False),
15     ("aba", True),
16 ]
17 for sentence, expected in test_cases:
18     result = is_sentence_palindrome(sentence)
19     print(f"'{sentence}' -> {result} (expected {expected})")
20     assert result == expected
21 print("All tests passed")

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

PS C:\Users\shash> c:: cd 'c:\Users\shash'; & 'c:\Users\shash\anaconda
-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '64904' '-- 'c:\U
'A man a plan a canal Panama' -> True (expected True)
'This is not a palindrome' -> False (expected False)
' ' -> True (expected True)
'a' -> True (expected True)
'A' -> True (expected True)
'ab' -> False (expected False)
'aba' -> True (expected True)
All tests passed
PS C:\Users\shash>

```

Task 4: ShoppingCart Class

Scenario

You are designing a basic shopping cart module for an e-commerce application.

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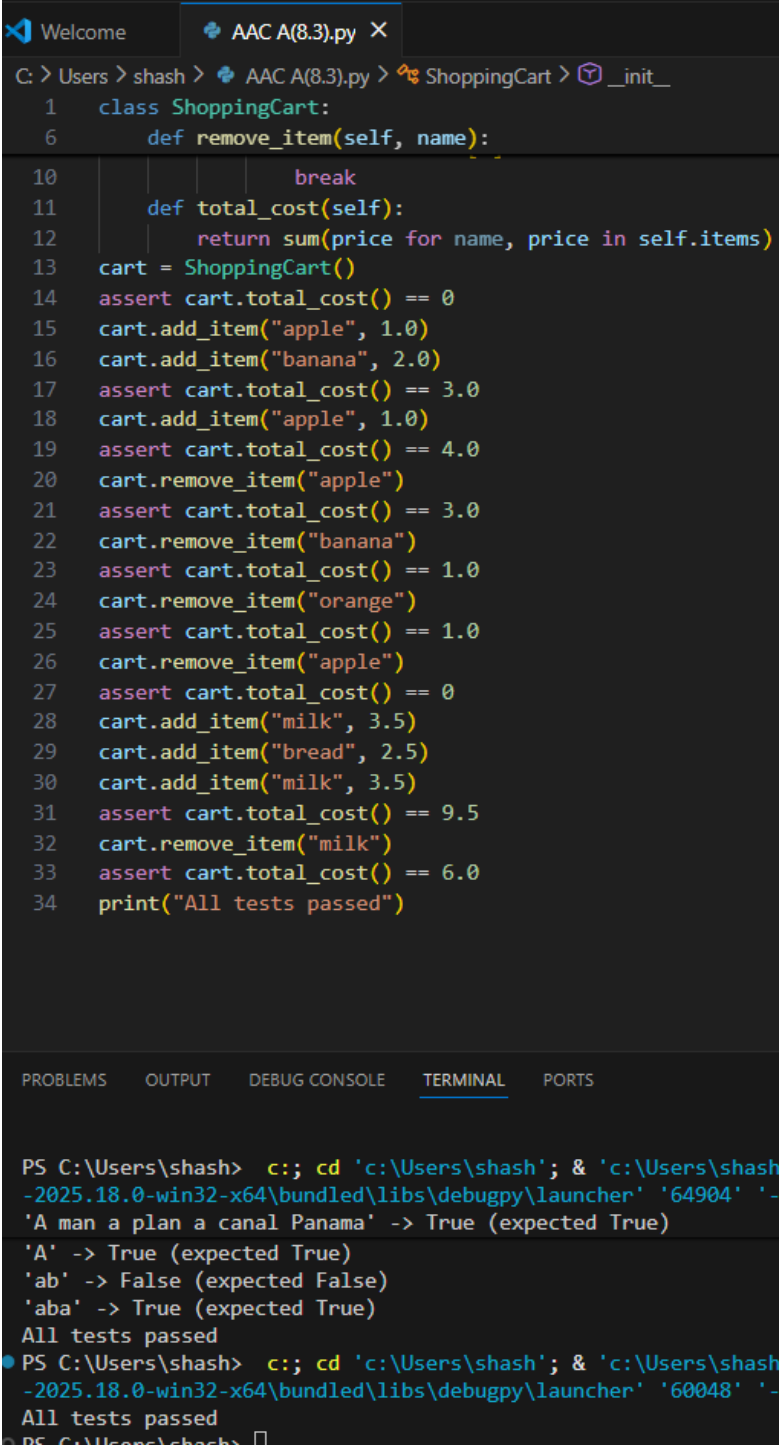
	Requirements • AI should generate test cases for the ShoppingCart class • Class must include the following methods: – add_item(name, price) – remove_item(name) – total_cost() • Validate correct addition, removal, and cost calculation • Handle empty cart scenarios Expected Output • Fully implemented ShoppingCart class • All methods pass AI-generated test cases • Total cost is calculated accurately • Items are added and removed correctly	
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```
Welcome AAC A(8.3).py X
C: > Users > shash > AAC A(8.3).py > ShoppingCart > _init_
1 class ShoppingCart:
2     def __init__(self):
3         self.items = []
4     def add_item(self, name, price):
5         self.items.append((name, price))
6     def remove_item(self, name):
7         for i, (n, p) in enumerate(self.items):
8             if n == name:
9                 del self.items[i]
10                break
11    def total_cost(self):
12        return sum(price for name, price in self.items)
13    cart = ShoppingCart()
14    assert cart.total_cost() == 0
15    cart.add_item("apple", 1.0)
16    cart.add_item("banana", 2.0)
17    assert cart.total_cost() == 3.0
18    cart.add_item("apple", 1.0)
19    assert cart.total_cost() == 4.0
20    cart.remove_item("apple")
21    assert cart.total_cost() == 3.0
22    cart.remove_item("banana")
23    assert cart.total_cost() == 1.0
24    cart.remove_item("orange")
25    assert cart.total_cost() == 1.0
26    cart.remove_item("apple")
27    assert cart.total_cost() == 0
28    cart.add_item("milk", 3.5)
29    cart.add_item("bread", 2.5)
30    cart.add_item("milk", 3.5)
31    assert cart.total_cost() == 9.5
32    cart.remove_item("milk")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

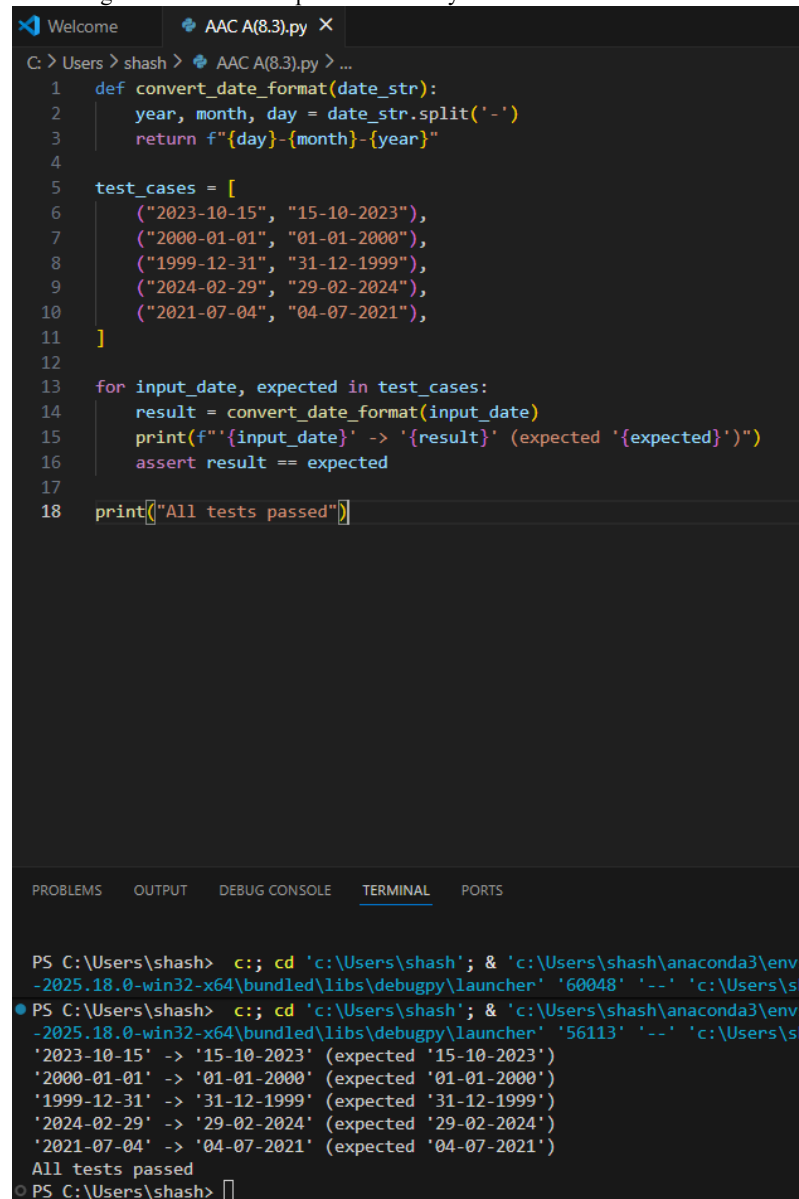
```
PS C:\Users\shash> c::; cd 'c:\Users\shash'; & 'c:\Users\shash\an
-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '64904' '--'
'A man a plan a canal Panama' -> True (expected True)
'A' -> True (expected True)
'ab' -> False (expected False)
'aba' -> True (expected True)
All tests passed
PS C:\Users\shash> c::; cd 'c:\Users\shash'; & 'c:\Users\shash\an
-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '60048' '--'
All tests passed
PS C:\Users\shash>
```

	 <pre> Welcome AAC A(8.3).py X C: > Users > shash > AAC A(8.3).py > ShoppingCart > _init_ 1 class ShoppingCart: 6 def remove_item(self, name): 10 break 11 def total_cost(self): 12 return sum(price for name, price in self.items) 13 cart = ShoppingCart() 14 assert cart.total_cost() == 0 15 cart.add_item("apple", 1.0) 16 cart.add_item("banana", 2.0) 17 assert cart.total_cost() == 3.0 18 cart.add_item("apple", 1.0) 19 assert cart.total_cost() == 4.0 20 cart.remove_item("apple") 21 assert cart.total_cost() == 3.0 22 cart.remove_item("banana") 23 assert cart.total_cost() == 1.0 24 cart.remove_item("orange") 25 assert cart.total_cost() == 1.0 26 cart.remove_item("apple") 27 assert cart.total_cost() == 0 28 cart.add_item("milk", 3.5) 29 cart.add_item("bread", 2.5) 30 cart.add_item("milk", 3.5) 31 assert cart.total_cost() == 9.5 32 cart.remove_item("milk") 33 assert cart.total_cost() == 6.0 34 print("All tests passed") PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS PS C:\Users\shash> c:; cd 'c:\Users\shash'; & 'c:\Users\shash -2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '64904' '- 'A man a plan a canal Panama' -> True (expected True) 'A' -> True (expected True) 'ab' -> False (expected False) 'aba' -> True (expected True) All tests passed ● PS C:\Users\shash> c:; cd 'c:\Users\shash'; & 'c:\Users\shash -2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '60048' '- All tests passed PS C:\Users\shash> </pre>	
	Task 5: Date Format Conversion Scenario You are creating a utility function to convert date formats for reports. Requirements - AI should generate test cases for <code>convert_date_format(date_str)</code> - Input format must be "YYYY-MM-DD" - Output format must be "DD-MM-YYYY" - Example:	

– "2023-10-15" → "15-10-2023"

Expected Output

- Date conversion function implemented in Python
- Correct format conversion for all valid inputs
- All AI-generated test cases pass successfully



```
1 def convert_date_format(date_str):
2     year, month, day = date_str.split('-')
3     return f"{day}-{month}-{year}"
4
5 test_cases = [
6     ("2023-10-15", "15-10-2023"),
7     ("2000-01-01", "01-01-2000"),
8     ("1999-12-31", "31-12-1999"),
9     ("2024-02-29", "29-02-2024"),
10    ("2021-07-04", "04-07-2021"),
11]
12
13 for input_date, expected in test_cases:
14     result = convert_date_format(input_date)
15     print(f'{input_date} -> {result} (expected {expected})')
16     assert result == expected
17
18 print("All tests passed")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\shash> c::; cd 'c:\Users\shash'; & 'c:\Users\shash\anaconda3\envs\
-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '60048' '--' 'c:\Users\sh
● PS C:\Users\shash> c::; cd 'c:\Users\shash'; & 'c:\Users\shash\anaconda3\envs\
-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '56113' '--' 'c:\Users\sh
'2023-10-15' -> '15-10-2023' (expected '15-10-2023')
'2000-01-01' -> '01-01-2000' (expected '01-01-2000')
'1999-12-31' -> '31-12-1999' (expected '31-12-1999')
'2024-02-29' -> '29-02-2024' (expected '29-02-2024')
'2021-07-04' -> '04-07-2021' (expected '04-07-2021')
All tests passed
PS C:\Users\shash> █
```

Note: Report should be submitted as a word document for all tasks in a single document with prompts, comments & code explanation, and output and if required, screenshots.